Model Quality Comparison: GPT-4o vs GPT-5-chat vs Copilot Studio Orchestration

NC Governor's Communications Office AI Agent — April 2026

Summary

We tested 7 questions across 4 configurations to answer one question: **does the quality survive the journey from our backend through Copilot Studio to the user?**

Configuration	Avg Score	Strengths	Gaps
Raw API — GPT-4o	**4.0 / 4**	Structured data, citations, themes	Solid baseline; GPT-5-chat is richer on remarks
Raw API — GPT-5-chat	**4.0 / 4**	Deeper analysis, evolution tables, more granular output	Same clips retrieval as GPT-4o (search doesn't use chat model)
Copilot Studio — Generative	**3.9 / 4**	Blends data sources, preserves quotes and citations	Proofread lost the itemized change list (3/4); corrections were accurate
Copilot Studio — Classic	3.2 / 4	High fidelity when routed correctly (Q7 scored 4/4)	Clips returned as raw JSON; remarks sometimes needed rephrasing

Bottom line: GPT-5-chat on the backend is a clear quality win for remarks synthesis — richer structure, deeper quote analysis, evolution tracking. The Copilot Studio GPT-4.0 orchestrator preserves this quality faithfully; direct quotes, citations, and thematic structure all survive the "last mile." The generative orchestration mode adds value by blending data sources into unified answers.

Test Configurations	
1. Raw API — GPT-4o	Azure Functions backend calling **GPT-4o** directly via APIM. text-embedding-3-large for vectors.

2. Raw API — GPT-5-chat	Same backend swapped to **GPT-5-chat** (gpt-5-chat, 2025-10-03). Embedding model unchanged.
3. Copilot Studio Generative	GCC environment, **GPT-4.0 orchestrator**. Generative orchestration calls the GPT-5-chat backend as tools, then re-summarizes for the user. Tests the "last mile" quality through the orchestration layer.
4. Copilot Studio Classic	GCC environment, **GPT-4.0 orchestrator**. Classic topics call the same backend. Author controls which fields appear in the response template. Tests whether manual topic design improves fidelity.

Grading Criteria

Each response is scored on four criteria. The score (out of 4) reflects how many criteria were met. A one-line justification explains the rating.

Criteria	What we looked for
Relevance	On-topic results that address the question asked
Citations	Clear source attribution — date, event, outlet, or document reference
Completeness	Covered the breadth of available data, not just a partial slice
Presentation	Well-structured, readable output appropriate for the audience

Q1: Clips — "What's the latest news about Hurricane Helene recovery?"

Capability: News Clips Search | **Type:** Clips query with web search + gov scraper sources

RAW API — GPT-4O

4/4

Relevant, cited (outlet/date/URL), complete, well-structured JSON

Returned 3 clips with structured data. Top result from ncbudget.org (2026-03-30) with full article text about unmet needs. Also qcnews.com and naswnc.org.

Response includes id, url, outlet, title, publishedAt, lede, mentionContext, mentionOffset, fullText for each clip.

RAW API — GPT-5-CHAT

4/4

Identical retrieval; clips search does not use the chat model

Same search results. Clips query does not use the chat model for retrieval, only for embeddings. Output identical to GPT-4o.

COPILOT STUDIO — GENERATIVE

4/4

Relevant, cited (7 outlets), complete (blended clips + remarks), synthesized bullets

Synthesized 5 bullet points:
- Funding/Grants ($24M mitigation grants)
- Federal Reimbursement ($155M)
- Congressional Request ($13.5B)
- Temporary Housing Extension (FEMA 6-month)
- State Budget ($792M)

Included footnotes citing NC Budget, ABC11, Spectrum News, WFMY, NC Newsline, NC Governor, Hoodline.

Blended clips + remarks data into one coherent answer.

COPILOT STUDIO — CLASSIC

3/4

Relevant, cited, complete (10 clips), but raw JSON — not formatted for chat

Returned "I found 10 on that" followed by raw JSON array of all 10 clips with fullText.

All data present but unformatted. User would need Adaptive Cards or message formatting to make it presentable. The data is complete — formatting is a topic design choice, not a quality issue.

Q2: Remarks — "What language has the Governor used about clean energy and economic development?"

Capability: Remarks Search (RAG synthesis) | **Type:** Hybrid search + GPT synthesis over Governor's speeches

RAW API — GPT-4O

4/4

Relevant, cited (date/event), complete (7 quotes, 5 themes), well-structured synthesis

5 themed sections, 7 direct quotes with (date, event) citations. "Evolution in Messaging" section tracking March 2025 vs November 2025.

Sources: 10 chunks from State of the State and November Column.

Key quotes:
- "We've become a leader in the clean energy economy"
- "more than 30,000 good-paying jobs"
- "more than 1500 new jobs, most of which are in rural North Carolina"

RAW API — GPT-5-CHAT

4/4

All criteria met; richer than GPT-4o — evolution table, deeper contextual analysis

Added markdown table tracking messaging evolution by period. More structured sections with explicit analysis of how language shifted.

Same 7+ quotes but with deeper contextual analysis of each. Explicitly noted where clean energy is "implicit rather than directly invoked" in later remarks.

COPILOT STUDIO — GENERATIVE

4/4

Relevant, cited, complete (4 themes), well-presented; quotes and citations preserved through orchestration

Good synthesis — 4 themed sections:
1. Core Framing
2. Integration of Clean Energy
3. Economic Development Themes
4. Evolution of Messaging

Preserved key quotes: "We've become a leader in the clean energy economy," "more than 30,000 good-paying jobs," "more than 1500 new jobs, most of which are in rural North Carolina."

Evolution section present. Slightly less granular than GPT-5-chat direct output but quotes and citations intact.

COPILOT STUDIO — CLASSIC

3/4

Relevant, cited, complete, but required rephrasing to route — presentation depends on trigger match

Strong synthesis with dual framing analysis:
1. **Clean Energy Framing** — "leader in the clean energy economy" [Ex. 1] as strategic economic leadership; "clean air to breathe and clean water to drink" [Ex. 4] as quality-of-life infrastructure.
2. **Economic Development Framing** — Competitiveness, rural inclusion, workforce/skills focus. Key quotes: "real shot at success," "tens of thousands of jobs," "most of new jobs are in rural North Carolina."
3. **Evolution Over Time** — March 2025 policy-driven competitiveness to December 2025 moral/civic "sacred responsibility to be a light." Noted clean energy not directly mentioned in later remarks but connected thematically.
4. **Summary** — "Clean energy and economic development as mutually reinforcing pillars of North Carolina's identity."

Direct quotes with [Ex. #] citations throughout. Content quality comparable to GPT-5-

			chat direct output. Note: Initial attempt failed ("could not answer") due to trigger phrase mismatch. Succeeded after rephrasing prompt.

Q3: Proofread — "The governer said North Carolinas econimic future is bright..."

Capability: Transcript Proofreading | **Type:** AI-powered spelling/grammar correction with itemized change list

RAW API — GPT-4O	RAW API — GPT-5-CHAT	COPILOT STUDIO — GENERATIVE	COPILOT STUDIO — CLASSIC
4/4	4/4	3/4	N/A
All errors caught, itemized changes with summary, well-structured output	*All criteria met; more granular than GPT-4o — individual items with confidence, caught "Governor" capitalization*	*Relevant and correct, but lost the itemized change list — just corrected text and a brief summary*	
Corrected text returned. 1 bulk change block listing all 5 corrections together. Changes: - governer -> governor - North Carolinas -> North Carolina's - econimic -> economic - dubble -> double - apprentises -> apprentices - Added comma before "and" Summary: "Corrected multiple spelling errors, added a possessive apostrophe, and inserted a comma."	Corrected text returned. 5 individual change items, each with its own confidence and reason. Also capitalized "Governor" (GPT-4o left it lowercase). More granular and structured output than GPT-4o.	Returned corrected text with one-line summary: "Capitalization and minor punctuation corrections were made for clarity." Lost the itemized change list entirely — just the corrected text and a brief note. The corrections themselves were accurate.	

Q4: Clips — "Has the Governor made any announcements about jobs or economic development?"

Capability: News Clips Search | **Type:** Clips query covering jobs/econ topics

RAW API — GPT-4O

4/4

Relevant, cited (outlet/date/URL), complete, well-structured JSON

3 clips returned:
- TSEA Energy facility in Eden (160 jobs)
- Capital Group Charlotte hub
- PSA Airlines HQ grand opening

Structured data with outlet, date, lede, URL.

RAW API — GPT-5-CHAT

4/4

Identical retrieval; clips search does not use the chat model

Same search results. Retrieval does not depend on the chat model.

COPILOT STUDIO — GENERATIVE

4/4

Relevant, cited (footnotes), complete (blended clips + remarks), 7 themed sections

Rich narrative response with 7 themed sections:
1. Workforce Development
2. Job Creation
3. TSEA Energy
4. Citi Charlotte
5. Child Care Task Force
6. Tax Policy
7. Recognition of Contributors

Blended clips and remarks data. Included footnotes. Went beyond listing clips to synthesize across both data sources.

COPILOT STUDIO — CLASSIC

3/4

Relevant, cited, complete (10 clips), but raw JSON — not formatted for chat

"I found 10 on that" + raw JSON. All data present, needs formatting.

Q5: Clips — "What's happening with Medicaid in North Carolina?"

Capability: News Clips Search | **Type:** Clips query on healthcare policy

RAW API — GPT-4O

4/4

Relevant, cited (outlet/date/URL), complete, well-structured JSON

3 clips:
- Carolina Journal (Stein urges Medicaid rebase funding)
- NC Newsline (Gov warns looming cuts)
- Newsline (Stein calls for special session)

Structured data with outlet, date, lede, URL.

RAW API — GPT-5-CHAT

4/4

Identical retrieval; clips search does not use the chat model

Same search results.

COPILOT STUDIO — GENERATIVE

4/4

Relevant, cited (6 outlets), complete (blended clips + remarks), 7-point narrative

Comprehensive 7-point narrative:
1. Funding Issues
2. Reimbursement Rate Restorations
3. Court Rulings
4. Political Pressure
5. Legislative Actions
6. Impact on Providers
7. Governor's Remarks

Cited WRAL, HME News, WUNC, NC Newsline, NC Health News, Carolina Journal. Blended clips + remarks for a complete picture.

COPILOT STUDIO — CLASSIC

3/4

Relevant, cited, complete (10 clips), but raw JSON — not formatted for chat

"I found 10 on that" + raw JSON. All data present.

Q6: Remarks — "What has the Governor said about public safety and law enforcement?"

Capability: Remarks Search (RAG synthesis) | **Type:** Hybrid search + GPT synthesis on public safety themes

RAW API — GPT-4O

4/4

Relevant, cited (date/event), complete (7+ quotes, 5 themes), well-structured with evolution tracking

Structured synthesis with themes:
- Support for law enforcement
- Violence prevention
- Mental health
- Gun safety
- Reentry

7+ direct quotes with date/event citations. Evolution tracked from March (salary focus) to October (holistic approach).

RAW API — GPT-5-CHAT

4/4

All criteria met; richer than GPT-4o — more granular quote analysis, deeper evolution tracking

Same themes plus more granular quote analysis. Bold section headers, explicit contextual analysis of each quote's significance.

Evolution tracked with more detail than GPT-4o.

COPILOT STUDIO — GENERATIVE

4/4

Relevant, cited, complete (6 themes), all key quotes preserved through orchestration

6 themed sections:
1. Support for Law Enforcement
2. Violence Prevention
3. Mental Health
4. Gun Safety
5. Reentry Support
6. Acknowledgment

All key quotes preserved with citations. Evolution section present (early = salary focus, October = systemic issues).

Content quality comparable to direct API output.

COPILOT STUDIO — CLASSIC

3/4

Relevant, cited, complete, but raw chunks mixed with synthesis — cluttered presentation

Raw JSON source chunks displayed, then synthesis below. User saw chunks + synthesis together.

Data was complete. Synthesis quality not separately visible from raw data in this test.

Q7: Remarks — "What has the Governor said about education and teacher pay?"

Capability: Remarks Search (RAG synthesis) | **Type:** Hybrid search + GPT synthesis on education policy

RAW API — GPT-4O

4/4

Relevant, cited (date/event), complete (6+ quotes), well-structured synthesis

Synthesis with direct quotes:
- "second-lowest in the southeast"
- "real pay raise"
- Workforce development connection

6+ quotes with citations. Evolution noted.

RAW API — GPT-5-CHAT

4/4

All criteria met; richer than GPT-4o — evolution table, education-to-workforce pipeline analysis

Markdown evolution table, more granular quote-by-quote analysis, explicit connection between education and workforce pipeline.

COPILOT STUDIO — GENERATIVE

4/4

Relevant, cited (dates), complete (3 sections + evolution), key quotes preserved

Good synthesis — 3 sections:
1. Low Starting Pay
2. Real Pay Raise
3. Teachers as Inspirational Figures

Plus Broader Context and Evolution sections.

Key quotes preserved: "second-lowest in the southeast, lower than every one of our bordering states — even South Carolina," "we must give teachers a real pay raise!"

Citations with dates. Evolution tracked.

COPILOT STUDIO — CLASSIC

4/4

Relevant, cited ([#] numbered), complete (5+ quotes), evolution tracked — strong synthesis

Synthesis shown (after raw JSON chunks). 5+ quotes with [#] numbered citations and date/event.

Tracked evolution from March policy-driven "real pay raise" to December values-based "miracle of light" framing. Summary paragraph ties it together.

Content quality comparable to or better than generative mode.

Scorecard

GPT-4o and GPT-5-chat both score 4/4 across the board — the difference is in **depth, not correctness**. GPT-5-chat produces richer synthesis on remarks: evolution tables, deeper contextual analysis, more structured sections. For clips, retrieval is identical since search doesn't use the chat model.

Question	Raw API — GPT-4o	Raw API — GPT-5-chat	Copilot Studio Generative	Copilot Studio Classic

Q1: Helene recovery (clips)	4/4 — 3 clips, structured	4/4 — identical retrieval	4/4 — 5 bullets, blended sources	3/4 — 10 clips, raw JSON
Q2: Clean energy (remarks)	4/4 — 7 quotes, 5 sections	4/4 — **richer:** evolution table, deeper analysis	4/4 — 4 sections, quotes preserved	3/4 — strong synthesis, needed rephrase
Q3: Proofread	4/4 — 5 corrections, itemized	4/4 — **more granular:** individual items + "Governor"	3/4 — correct, but lost change list	N/A
Q4: Jobs/econ (clips)	4/4 — 3 clips, structured	4/4 — identical retrieval	4/4 — 7 sections, blended sources	3/4 — 10 clips, raw JSON
Q5: Medicaid (clips)	4/4 — 3 clips, structured	4/4 — identical retrieval	4/4 — 7-point narrative, 6 outlets	3/4 — 10 clips, raw JSON
Q6: Public safety (remarks)	4/4 — 7+ quotes, themes	4/4 — **richer:** granular quote analysis	4/4 — 6 sections, quotes preserved	3/4 — chunks + synthesis mixed
Q7: Education (remarks)	4/4 — 6+ quotes, evolution	4/4 — **richer:** evolution table, pipeline analysis	4/4 — 3 sections + evolution	4/4 — 5+ quotes, numbered citations
Average	**4.0**	**4.0**	**3.9**	**3.2 (6 tested)**

Key Findings

1. GPT-5-chat produces richer synthesis than GPT-4o.

The upgrade is most visible in remarks synthesis: more structured sections, markdown tables for evolution tracking, deeper contextual analysis of individual quotes. For clips queries, the difference is negligible since retrieval does not depend on the chat model. The GPT-5-chat backend is a clear upgrade for the highest-value use case (remarks analysis).

2. Generative orchestration preserves content quality well.

Direct quotes, citations, thematic structure, and evolution tracking all survive the GPT-4.0 "last mile." The orchestrator adds value by blending clips + remarks into unified answers (Q1, Q4, Q5). Scored 4/4 on all clips and remarks questions. Only gap: the proofread change list gets flattened to a single summary (3/4 on Q3). For remarks synthesis — the highest-value use case — quality degradation is minimal.

3. Classic orchestration has higher fidelity when it works, but reliability is a problem.

The synthesis text comes through with full quotes and numbered citations (Q7, Q2 were both excellent). Q2 required rephrasing to match the topic trigger, exposing a sensitivity to prompt wording. Classic gives you control over what fields to display but requires careful trigger phrase engineering. Once routed correctly, content quality is comparable to the direct API output.

4. The "last mile" problem is smaller than expected.

The concern was that Copilot Studio's GPT-4.0 orchestrator would degrade the backend output. For remarks synthesis (the highest-value use case), the quality degradation is minimal. Quotes are preserved, citations are maintained, thematic structure survives. The GPT-5-chat backend does the heavy analytical lifting; Copilot Studio presents it faithfully.